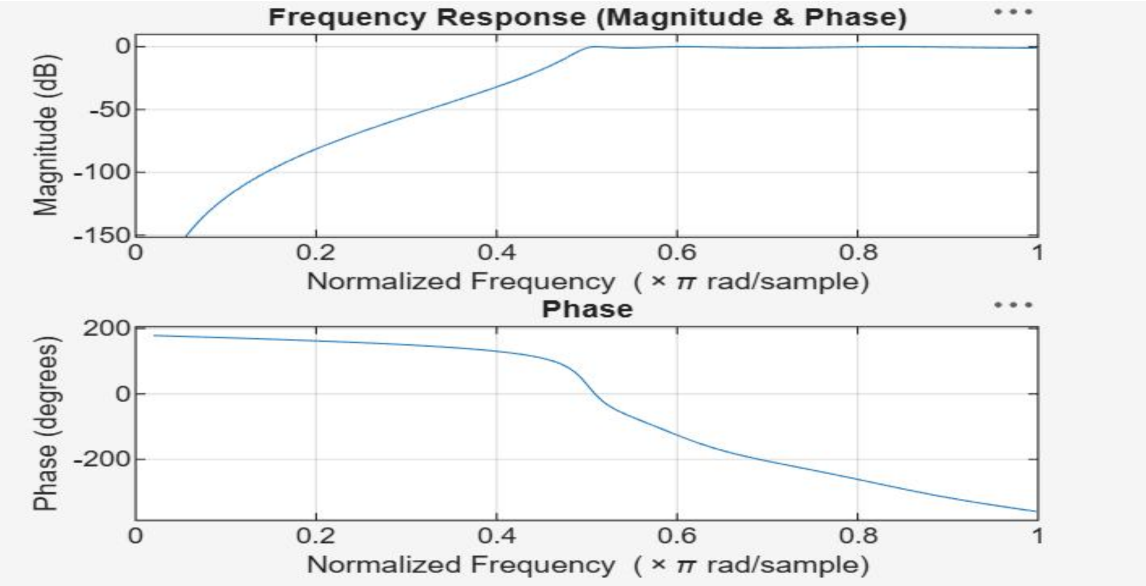
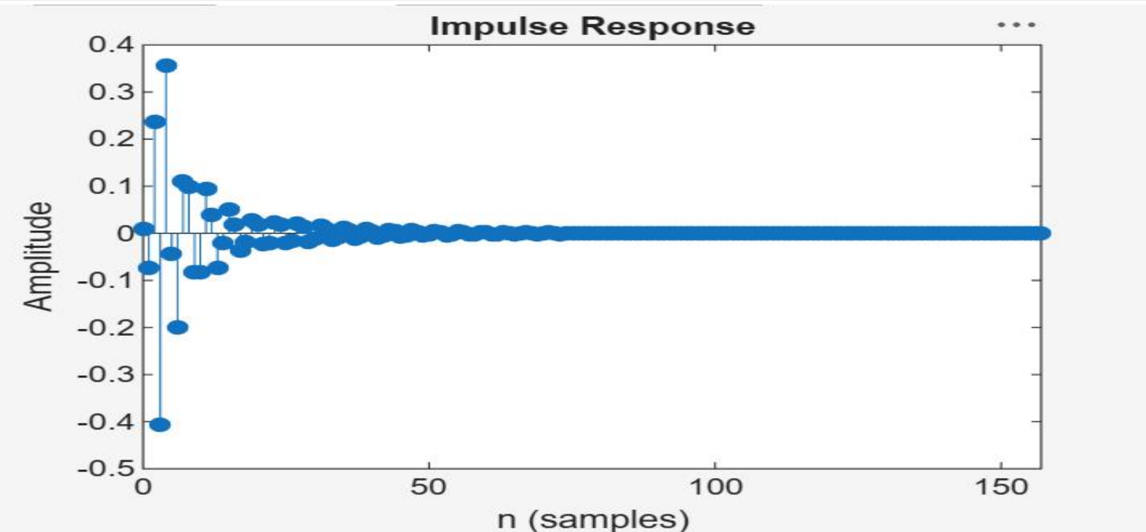
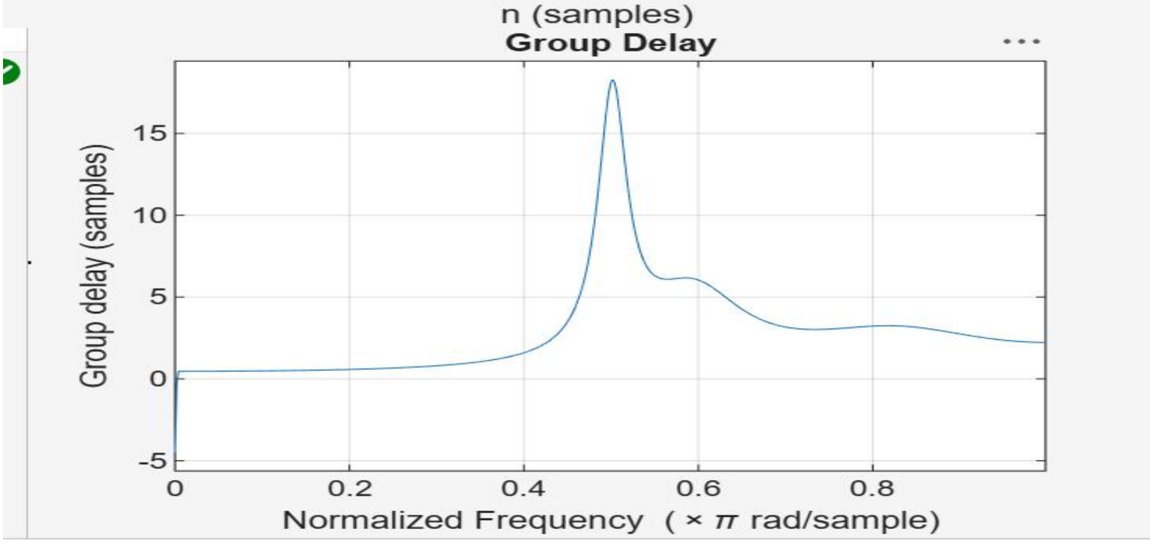
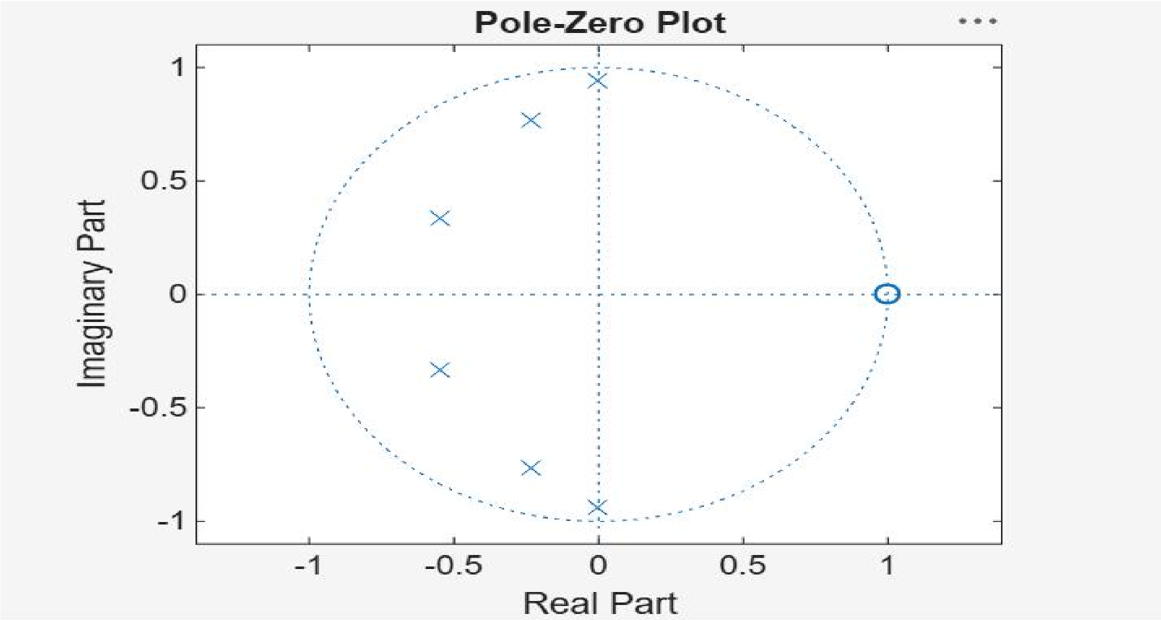
EXP 2 (2.1)

```
clc; clear;

% EX NO:2
% Chebyshev IIR Filter
% DATE:
% 2.1. Design and Performance Evaluation of a Chebyshev Type-I IIR Low-Pass
Filter

Wp = 0.4; % Passband edge (normalized, 0 to 1 → π rad/sample)
Ws = 0.55; % Stopband edge (normalized)
Rp = 1; % Passband ripple (dB)
As = 40; % Stopband attenuation (dB)

% Filter order and cutoff frequency
[N, Wn] = cheb1ord(Wp, Ws, Rp, As);
[b, a] = cheby1(N, Rp, Wn);

% Display results
fprintf('Filter Order = %d\n', N);
fprintf('Cutoff Frequency = %.2f * pi rad/sample\n', Wn);

% Stability check
if all(abs(roots(a)) < 1)
disp('System is STABLE');
else
disp('System is UNSTABLE');
end

% -------- GRAPHS --------
% Magnitude & Phase Response
figure;
freqz(b, a);
title('Magnitude and Phase Response');

% Impulse Response
figure;
impz(b, a);
title('Impulse Response');

% Group Delay
figure;
grpdelay(b, a);
title('Group Delay');

% Pole-Zero Plot
figure;
zplane(b, a);
title('Pole-Zero Plot');
```


Frequency Response (Magnitude & Phase)
Magnitude (dB)
0
-50
-100
-150
0
0.2
0.4
0.6
0.8
1
Normalized Frequency (× π rad/sample)
Phase
Phase (degrees)
200
0
-200
0
0.2
0.4
0.6
0.8
1
Normalized Frequency (× π rad/sample)


Impulse Response
Amplitude
0.4
0.3
0.2
0.1
0
-0.1
-0.2
-0.3
-0.4
-0.5
0
50
100
150
n (samples)


Group Delay
Group delay (samples)
15
10
5
0
-5
0
0.2
0.4
0.6
0.8
Normalized Frequency (× π rad/sample)

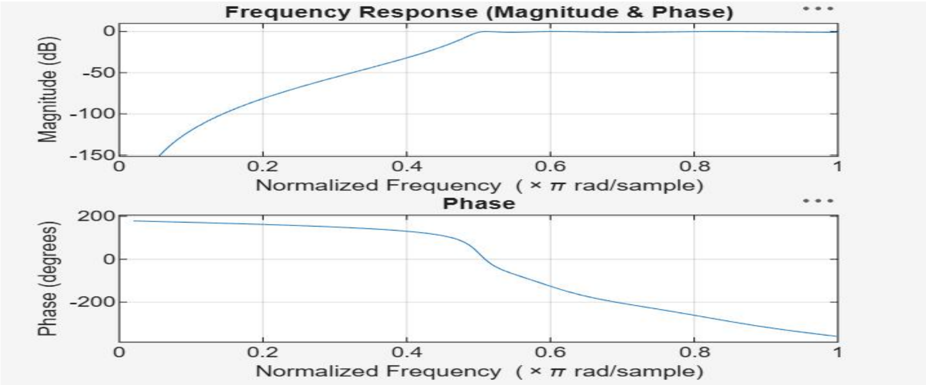


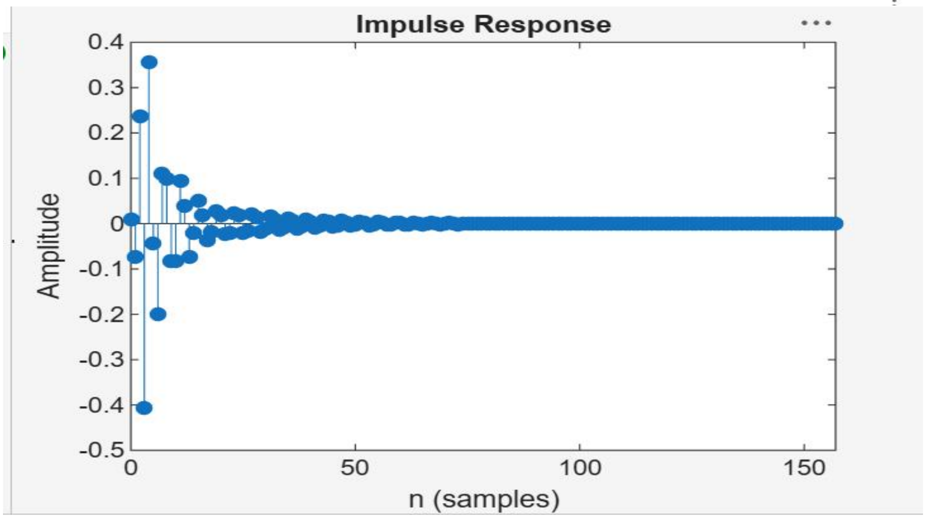

```
Filter Order = 6
Cutoff Frequency (Wn) = 0.5000 (normalized)
>>
```

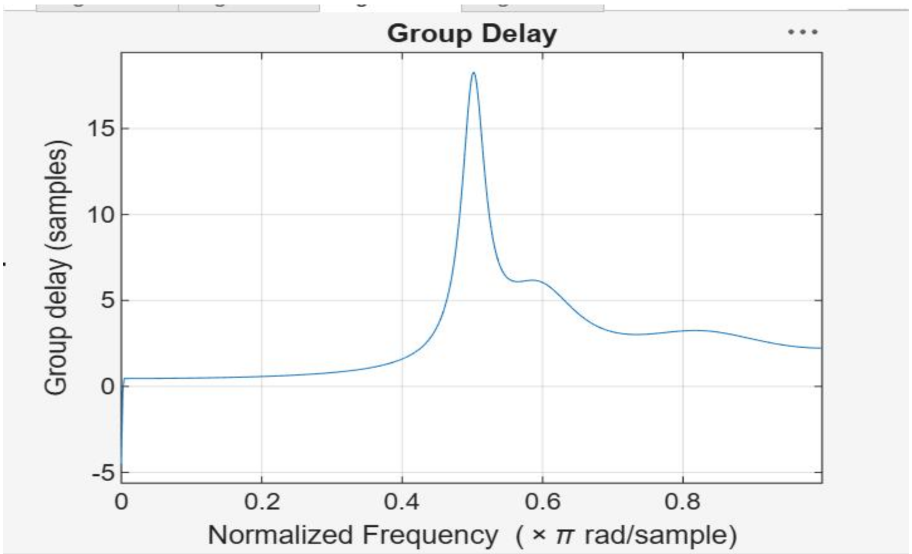
EXP-2 (2.2)

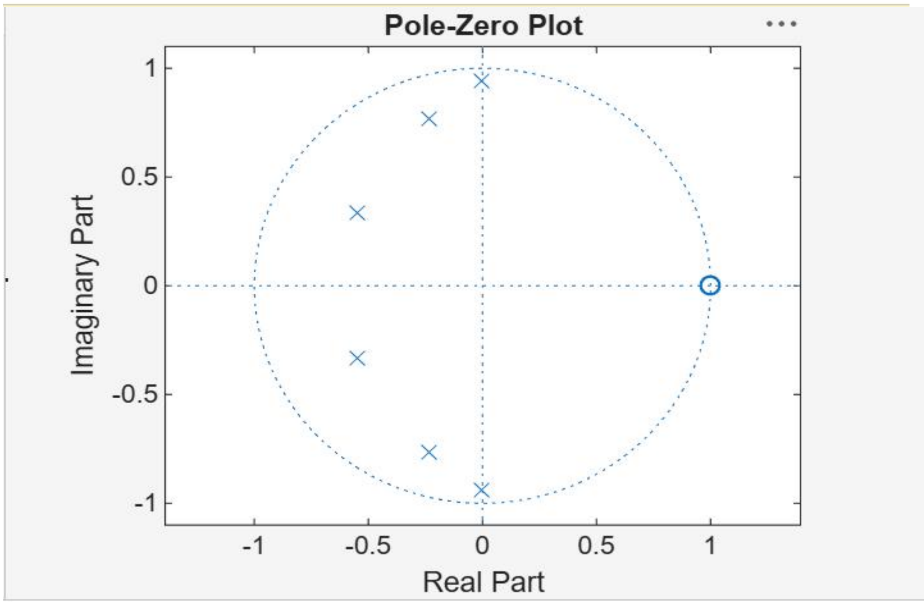
```
clc;
clear;
close all;

% EX NO:2
% Chebyshev IIR Filter
% DATE:
% 2.2. Design and Performance Evaluation of a Chebyshev Type-I IIR High-
Pass Filter

% Specifications
Rp = 1; % Passband Ripple (dB)
Rs = 40; % Stopband Attenuation (dB)
Wp = 0.5; % Passband Edge (normalized, 0 to 1 → π rad/sample)
Ws = 0.35; % Stopband Edge (normalized)

% Find Minimum Filter Order
[n, Wn] = cheb1ord(Wp, Ws, Rp, Rs);

% Design High-Pass Chebyshev Type-I Filter
[b, a] = cheby1(n, Rp, Wn, 'high');

% Display Results
fprintf('Filter Order = %d\n', n);
fprintf('Cutoff Frequency (Wn) = %.4f (normalized)\n', Wn);

% -------- GRAPHS --------
% Frequency Response
figure;
```

```
freqz(b, a);
title('Frequency Response (Magnitude & Phase)');

% Impulse Response
figure;
impz(b, a);
title('Impulse Response');

% Group Delay
figure;
grpdelay(b, a);
title('Group Delay');

% Pole-Zero Plot
figure;
zplane(b, a);
title('Pole-Zero Plot');
```










```
Filter Order = 6
Cutoff Frequency (Wn) = 0.5000 (normalized)
>>
```